

**Communicative Efficiency in Adjective Ordering: Plan-Guided Production
between Hierarchical Semantic Composition and Incremental Pragmatic Choice**

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Abstract

Adjective ordering must reconcile stable order expectations with the usefulness of properties in the current scene. Theoretical accounts locate communicative efficiency in two directions: hierarchical semantic composition interprets noun-adjacent modifiers before outer modifiers, whereas surface production makes linearly earlier information available first. We test how these pressures are expressed in explicit order judgements and in the full distribution of produced descriptions. Two German slider studies show a replicated context gradient and a small residual size-first preference when colour alone identifies the target. A production study shows that speakers often resolve reference with one adjective, so adjective selection and utterance length can pre-empt an ordering decision. We compare matched Rational Speech Act models over fifteen completed responses. After controlling participant-specific variation in language-model-derived stable order expectations, a model with a shared interior prefix-normalisation weight predicts responses better than globally normalised and fully prefix-normalised endpoints ($\kappa = .447$, 95% interval $[.404, .490]$). Its advantage spans adjective selection, length, adjective set, and initial adjective. Belief-dependent size updating adds no predictive value in the fitted data; controlled simulation shows only that such updating can generate an asymmetry under stronger semantic assumptions. The results support a graded completed-response architecture, not a direct identification of token-by-token planning.

Keywords: adjective ordering, reference production, Rational Speech Act, prefix normalisation, Bayesian cognitive modelling

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1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with scene-specific informativeness in referential communication. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Scontras, 2023; Sproat & Shih, 1991). Yet the first adjective is also the listener’s first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica & Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

But what counts as efficient—and therefore early—for the listener in a complex referential setting? One relevant line of work concerns adjective subjectivity. More subjective adjectives tend to precede less subjective adjectives in prenominal sequences (Scontras, Degen & Goodman, 2017). Some subjectivity-based proposals connect this tendency to hierarchical semantic composition, under which the noun-adjacent adjective composes before adjectives farther from the noun. In *the big blue sticker*, the noun-adjacent adjective *blue* therefore combines with *sticker* before the linearly earlier *big*, narrowing the comparison class against which the gradable size adjective is interpreted. Because *blue* is typically less subjective than *big*, its noun-adjacent position may provide a comparatively stable restriction before the size adjective is interpreted, potentially contributing to the surface size-first preference (Franke, Scontras & Simonic, 2019; Scontras, Bar-Sever, Kachakeche, Rosales Jr & Samonte, 2020; Scontras, Degen & Goodman, 2019). A separate line of referential work suggests that adjective order is also sensitive to the discriminatory strength of each property in the current scene (Fukumura, 2018; Rubio-Fernández et al., 2021). When colour distinguishes the target more effectively than size, colour-first orders may become more

acceptable or more likely, consistent with a pressure to provide contextually useful information earlier in the surface sequence.

Depending on which perspective is taken, communicative efficiency may support conflicting adjective-ordering predictions. This tension is especially consequential in reference games (Frank & Goodman, 2012; Franke, 2009), where successful communication depends both on the completed description and on how information becomes available as that description unfolds. Work on incremental language processing emphasizes the latter: listeners integrate partial linguistic input with visual context to restrict candidate referents before the noun phrase is complete (Sedivy, Tanenhaus, Chambers & Carlson, 1999; Tanenhaus, Spivey-Knowlton, Eberhard & Sedivy, 1995). On this view, placing the most discriminating modifier early in the linear sequence may facilitate identification before the remaining words are heard (Rubio-Fernandez & Jara-Ettinger, 2020). On the other hand, the subjectivity-based account emphasizes the former by evaluating a completed adjective sequence through the listener’s hierarchical interpretation. From this comprehension-oriented perspective, an order is efficient when the noun-adjacent adjective provides a more reliable restriction for interpreting the outer adjective (Franke et al., 2019; Scontras et al., 2020, 2019). At the level of production, these two directions may guide different parts of the same process. A representation of the intended, hierarchically interpreted description can support an advance plan. Scene-specific usefulness can continue to shape choices among the remaining adjectives as the description unfolds. We refer to this coordination as *plan-guided incremental production*: speakers begin with a partial plan for the description and remain responsive to local alternatives during its surface realisation.

Moreover, before a speaker can weigh alternative adjective orders, they first decide which properties to mention and when to stop. In a colour-sufficient scene, *the blue sticker* may complete the description, so competition between size–colour and colour–size orders may never arise in the first place. Previous empirical studies show that colour is included redundantly more often than relative adjectives such as size (Rubio-Fernández, 2016;

Tarenskeen, Broersma & Geurts, 2015). Such redundancy can still support efficient identification, especially when modifier meanings are uncertain and gradable, as with size adjectives (Degen, Hawkins, Graf, Kreiss & Goodman, 2020; Rubio-Fernández, 2016). Scene-specific usefulness may therefore affect whether an adjective is used redundantly and, conditional on such redundant use, how the adjectives are ordered within a multi-adjective string.

Taken together, these considerations place adjective order within a broader production problem. Under plan-guided incremental production, advance commitment and local adjustment can influence which properties are selected, how many adjectives are produced, and how jointly selected adjectives are ordered. The relevant empirical target is therefore the full distribution of one-, two-, and three-adjective strings, with adjective order treated as a conditional outcome of adjective selection and utterance length. We ask three questions, moving from the empirical patterns to computational accounts of the mechanisms that may generate them. First, how do stable size-first preferences and scene-specific referential usefulness jointly shape explicit order judgements and the production outcomes of adjective selection, utterance length, and conditional order? Second, after stable order expectations are controlled, does the production distribution favour global evaluation of complete descriptions, fully incremental adjective choice, or plan-guided incremental production between these endpoints? Third, does sequential context updating within hierarchical semantic composition improve prediction in the present production data, and under which controlled conditions can it generate an order asymmetry?

Isolating the source of contributions of each account remains methodologically challenging because different mechanisms can generate the same surface distribution. We therefore combine carefully controlled behavioural experiments, computational cognitive modelling, simulations across a broader range of randomly generated contexts, and Bayesian predictive model comparison. The empirical component comprises two slider studies and one production study. Study 1a and its direct replication, Study 1b, measure explicit preferences

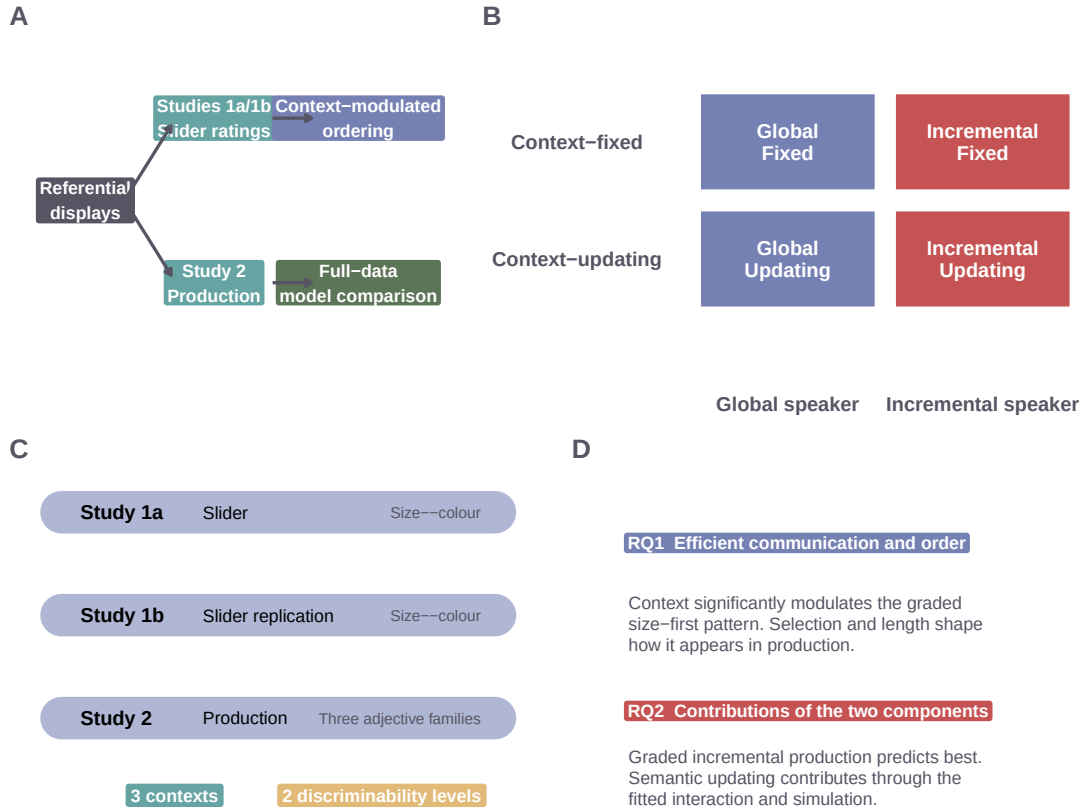
between size-first and colour-first descriptions across referential contexts using sliders. Study 2 uses direct production to elicit descriptions and records adjective selection, utterance length, and order conditional on producing the same adjective set. Referential context varies the relative usefulness of size and colour within each scene. Moreover, size discriminability is manipulated between participants, making size a more or less reliable visual cue throughout the experiment. Crossing these factors separates the relative usefulness of size in a particular scene from its perceptual reliability throughout the task. The former directly operationalises discriminatory strength. The latter provides a perceptual-reliability diagnostic relevant to the gradable size meaning invoked by subjectivity-based accounts, with lexical subjectivity held constant.

On the modelling side, we formalise these questions within the Rational Speech Act framework (Degen, 2023; Frank & Goodman, 2012; Goodman & Frank, 2016). All models assume hierarchical semantic composition. They differ in whether the interpretation of size remains anchored to the original scene or is recomputed after another adjective restricts the relevant context. They also differ in how descriptions are evaluated for production. A global speaker evaluates complete descriptions. An incremental speaker evaluates each adjective against the alternatives available at that point in the unfinished description. The graded model allows the competitive consequences of these successive choices to vary between the two endpoints. We interpret an interior value as plan-guided incremental production: an advance description-level plan constrains production, and competition among the remaining adjectives continues to shape its surface realisation. Our comparison allows the consequences of these successive evaluations to range from absent at the global endpoint to fully retained at the incremental endpoint. We estimate where along this range the observed production distribution is predicted best. All models predict the same distribution of fifteen possible completed responses. We compare their predictive accuracy using approximate leave-one-out prediction, with uncertainty assessed across participants. Separate simulations test when sequential semantic updating is sufficient to generate an order asymmetry.

Empirically, the slider studies reveal a replicated context gradient: size-first preferences are strongest when size identifies the target and weakest when colour does. A small residual size-first preference remains when colour alone identifies the target. The production study further shows that one-adjective descriptions often resolve reference before an ordering contrast arises. Lower size discriminability primarily increases the production of additional adjective material, especially in size-sufficient scenes. These findings place adjective order within a broader reference-generation problem in which scene-specific usefulness affects adjective selection and utterance length before an order among multiple adjectives becomes observable. After accounting for variation across participants in stable order preferences, predictive accuracy is highest at an intermediate point on the range from global to fully incremental evaluation. This pattern is consistent with plan-guided incremental production: advance commitment constrains the description, and local competition continues to shape its realisation. The predictive gain extends across adjective selection, utterance length, adjective set, and initial adjective. Belief-dependent size updating adds no meaningful predictive accuracy in the fitted production data. Controlled simulations identify conditions under which hierarchical updating has stronger semantic consequences and generates an order asymmetry.

The remainder of the paper is structured as follows. Section 2 preserves the two directional accounts of communicative efficiency and relates them to adjective selection and completed-response modelling. Section 3 reports the behavioural studies. Sections 4 and 5 introduce the production model and estimate where prediction is strongest along the global-to-incremental range. Section 6 evaluates the fitted and generative roles of semantic updating, and Section 7 considers the implications and limits of the findings.

Replace the overview figure with the final evidential pipeline and G-HO/K-HO/I-HO architecture.

**Figure 1**

Planned overview of the final evidential structure. The behavioural studies distinguish explicit order judgements from adjective selection and length in production. The production model separates stable order expectations and terminal response-policy terms from the global, graded, and incremental completed-response architectures. Semantic updating is evaluated as a separate predictive and simulation boundary question.

2 Background

Research on adjective ordering has identified stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), class-based hierarchies (Cinque, 1994; Dixon, 1982; Quirk, Greenbaum, Leech & Svartvik, 1972; Scott, 2002), and information-theoretic measures (Dyer, Torres, Scontras & Futrell, 2023; Futrell, Dyer & Scontras, 2020; Hahn, Degen, Goodman,

Jurafsky & Futrell, 2018). These predictors explain substantial structure, while referential context contributes additional variation that fixed hierarchies do not fully capture (Fukumura, 2018; Trotzke & Wittenberg, 2019). The present paper focuses on two directional accounts of communicative efficiency and on the production decisions that determine whether an order contrast arises.

2.1 Subjectivity, Hierarchical Composition, and Adjective Order

A central empirical generalisation in the adjective-ordering literature is the relation between an adjective’s *subjectivity* and its preferred distance from the noun. Scontras et al. (2017) operationalised subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences. Subsequent work suggests that the relation recurs across languages (Scontras, 2023; Trainin & Shetreet, 2021). It is directly relevant to size–colour sequences because size adjectives are typically more subjective and gradable than colour adjectives, yielding an expected size–first surface preference.

Scontras et al. (2019) locate the explanatory pressure in restrictive composition. Assuming a hierarchical prenominal structure, the adjective closest to the noun combines with the nominal denotation before an outer adjective applies. The outer, more subjective adjective can therefore be evaluated against a comparison class already restricted by the inner, less subjective adjective. Franke et al. (2019) make the same intuition explicit in simulation, showing that subjectivity-aligned orders can increase average model-assigned target probability under sequential composition with noisy meanings. This account assigns priority through the semantic hierarchy: in a surface size–colour sequence, colour composes before size. The proposal concerns formal dependency and leaves the real-time sequence of semantic processing open.

Scontras et al. (2020) develop this idea as *incremental semantic restriction*. Sequential narrowing of the comparison class can generate an ordering pressure independently of incremental pragmatic production in a process-model sense. For the present

materials, belief-dependent interpretation of size is therefore a plausible candidate source of a stable size-first asymmetry. The fitted analyses below test whether that candidate contributes predictively in the empirical regime; the simulation tests only whether it can generate an asymmetry under controlled assumptions.

2.2 Referential Usefulness, Adjective Selection, and Length

A second directional pressure comes from referential usefulness in the emerging surface sequence. Adjective use is sensitive to whether a modifier helps the listener identify the referent in the current scene (Degen et al., 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). This literature treats adjective choice and order as parts of a broader reference-generation problem. A related commitment appears in conceptualisation models that derive property choice from the contrast set available to the speaker (Fukumura, Van Gompel & Pickering, 2010; Van Gompel, Van Deemter, Gatt, Snoeren & Krahmer, 2019).

Fukumura (2018) provide a direct empirical precedent. Speakers were more likely to place the more discriminating adjective first, consistent with the proposal that order reflects which property best distinguishes the target in the current scene. Rubio-Fernández and colleagues develop this proposal as *incremental efficiency*: an expression can be useful because it narrows the listener’s candidate set before the description is complete (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). This production-side direction aligns with evidence that listeners integrate partial linguistic input with a visual context (Sedivy et al., 1999; Tanenhaus et al., 1995).

The same referential pressure can affect adjective selection and utterance length before it affects order. A scene may elicit one sufficient modifier, while perceptual or partner uncertainty can encourage additional, semantically redundant material (Degen et al., 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). An observable order contrast therefore depends on speakers selecting more than one adjective. This dependency motivates the full response-space analysis developed below.

2.3 Probability Architectures for Reference Generation

Psycholinguistic work shows that speakers can begin an utterance before later material is fully prepared and that planning scope varies across tasks and structures (Ferreira & Swets, 2002; Griffin & Bock, 2000; Konopka & Meyer, 2014). Those findings motivate process hypotheses, but completed descriptions do not by themselves reveal when particular words were planned. The Rational Speech Act framework instead provides a formal basis for comparing the distributional consequences of alternative choice architectures (Degen, 2023; Frank & Goodman, 2012; Goodman & Frank, 2016).

A globally normalised speaker assigns probability across completed descriptions. A prefix-normalised speaker evaluates each next adjective relative to a prefix-specific alternative set (Cohn-Gordon, Goodman & Potts, 2019). Both yield a final distribution over completed responses. The formal difference is whether the normalisers associated with successive alternative sets remain expressed in that completed-response distribution. Treating global and full prefix normalisation as endpoints allows a graded model to estimate the effective contribution of those normalisers without equating the parameter with planning time.

Existing incremental pragmatic models and models of redundant modification show that prefix-sensitive reasoning can capture position and order asymmetries (Degen et al., 2020; Rubio-Fernández et al., 2021). Schlotterbeck and Wang (2023) combine a related production architecture with belief-dependent interpretation of size in the slider paradigm used here. That account successfully motivates the present theoretical contrast, but it leaves open which components are required to predict the complete production response space. The final comparison therefore separates the completed-response normalisation architecture from stable order expectations, terminal selection and length terms, and the secondary semantic-updating contrast.

3 Behavioural Studies

The three studies expose different parts of the same empirical problem. Studies 1a and 1b ask participants to compare two completed size–colour orders directly. Study 2

instead elicits a description and therefore reveals whether speakers mention size, colour, or form, how many adjectives they produce, and only then how selected adjectives are ordered. The tasks can converge in their sensitivity to referential context without measuring the same behavioural construct.

The slider studies test whether a size-first preference is graded by referential usefulness and whether a small residual preference remains when colour alone identifies the referent. The size-discriminability manipulation supplies a secondary display-level diagnostic, but it is not treated as a direct manipulation of lexical subjectivity. The production study tests how context and visual size reliability affect adjective selection, utterance length, initial adjective, and order conditional on producing the same adjective set.

3.1 Shared Method

3.1.1 *Design*

The three studies shared the same basic referential design. The response format differed across experiments. The visual displays and target-definition logic were shared. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. The slider analyses focus on the theoretically central size–colour subset.¹ Within that subset, referential context varied within participants and had three levels: size-sufficient, colour-sufficient, and both-necessary. A second, between-participant factor, size discriminability, varied the size gap between the target and its size competitors. High size-discriminability displays used larger target–competitor gaps, making the size contrast visually pronounced. Low size-discriminability displays used smaller gaps, making size a less reliable cue. This factor was manipulated between participants so that each participant experienced a consistent level

¹ Study 2 also included size–form and colour–form trials, which enter the complete production model comparison. The size–colour subset remains the theoretical anchor because it provides the clearest contrast between a gradable adjective and a determinate feature adjective. The appendix reports the exact size values, target–competitor gap construction, and list assignment.

of visual size contrast throughout the task (Figure 2).

A

Design dimension	Levels
Referential context	First property sufficient; both necessary; second property sufficient
Size discriminability	High; low
Slider coverage	Size–colour
Production coverage	Size–colour; size–form; colour–form

B

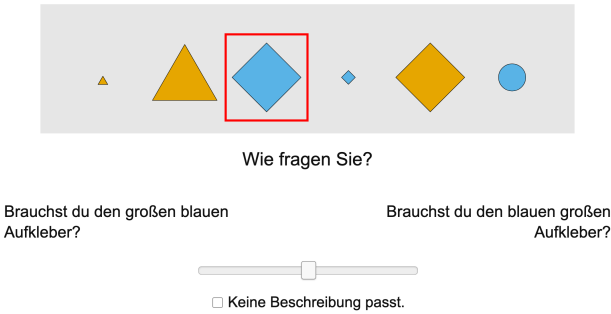


Figure 2

Experimental design and materials. Panel A summarises the shared design dimensions and task coverage. Panel B reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders.

3.1.2 Materials

Items were distributed across participants in a Latin-square design, so that scene templates and condition assignments were counterbalanced across lists. In addition to list counterbalancing, the critical scenes were realised through systematic permutations of colour and form values. This created variability in the surface displays and preserved the referential structure required by each condition. These permutations were applied to 27 critical scene templates, each specifying an abstract six-object configuration with a target, competitors, and distractors. Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene set; the modified noun was always *Aufkleber*

(‘sticker’).² The computational analysis reconstructed each display deterministically from the raw stimulus record as a 6×3 matrix, with one row per object and columns for observed size (*s*), colour (*c*), and form (*f*). The referent was stored at index 0. Participants saw the corresponding visual arrays; no stochastic feature recoding enters the final analysis.

3.1.3 Procedure

All studies were conducted online via Prolific. Participants in each experiment were instructed to identify the framed target using its perceptual properties. Spatial descriptions such as *left* or *right* were excluded. The task was framed as a simple reference game about collectible stickers: participants saw one framed target and gave or evaluated a description for a partner who saw the same stickers in a different order and lacked the target frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. Each study began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The experiment-specific response formats, trial counts, and response coding are reported below.

3.2 Study 1a: Slider-Rating Study

3.2.1 Participants

Participants ($N = 134$ after exclusions; 70 in the LOW size discriminability group, 64 in the HIGH size discriminability group) were recruited online via Prolific and reported German as their native language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three “none fits” responses on

² The form values were presented with German adjectival forms, such as *runden* for round/circle and *quadratischen* for square, so that form could occupy the same prenominal adjective position as colour and size.

experimental items.³ After exclusions, the analysed dataset contained 10,606 critical observations. The size–colour subset reported in the main empirical and model-comparison figures contained 3,485 observations; the appendix summarises exclusion details. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

3.2.2 Procedure

Study 1a implemented the shared referential design as a comparative judgment task. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a “none of the descriptions fits” option. Left–right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Responses were reoriented so that larger values favoured the size-first order and rescaled from 0–100 to 0–1. For the descriptive figure and linear mixed model, we then subtracted 0.5, yielding a centred scale from -0.5 to 0.5 with zero as the neutral point. The Bayesian predictive analysis retained the same size-first orientation; neutrality corresponds to the midpoint of the response scale.

3.2.3 Results

Figure 3 shows a clear referential-context gradient in size-first ratings. We analysed centred slider ratings with linear mixed-effects models fit in R using `lme4::lmer` (Bates, Mächler, Bolker & Walker, 2014). The reported model included referential context, size discriminability, and their interaction as fixed effects, together with participant intercepts and participant slopes for referential context. Effects were evaluated by likelihood-ratio

³ Control items were filler trials with an unambiguous expected response, included to screen for inattentive or non-referential responding. The additional “none fits” option allowed participants to indicate that neither description identified the target; frequent use of this option on experimental items was treated as evidence of a response strategy outside the intended comparative-judgment task.

comparisons between nested models, and planned marginal contrasts were computed with `emmeans`. We complemented these tests with a Bayesian predictive analysis using the same fixed effects and participant-level random-effects structure. The Bayesian analysis evaluates the planned hypotheses through predicted ratings for a new participant and reports posterior medians, 95% credible intervals, and directional posterior probabilities.

Referential context had a significant effect, $\chi^2(2) = 72.2$, $p < .001$. Mean centred ratings, indexing preference for size-first orderings, were highest in size-sufficient displays, lower in both-necessary displays, and lowest in colour-sufficient displays. Mean centred ratings remained above zero in colour-sufficient contexts, yielding no qualitative reversal. Planned contrasts preserved this gradient: size-sufficient ratings exceeded colour-sufficient ratings by 0.268, Bonferroni-corrected 95% CI [0.201, 0.335], and both-necessary ratings exceeded colour-sufficient ratings by 0.172, [0.126, 0.218]. The marginal colour-sufficient estimate was positive at 0.053, and its 95% CI narrowly included zero, $[-0.001, 0.106]$, $p = .056$. The Bayesian predictive analysis assigned posterior probability greater than .999 to the graded context order. It estimated a colour-sufficient preference of .054, 95% credible interval [.003, .109], with posterior probability .982 of exceeding the neutral point.

Size discriminability made a smaller contribution through a significant crossover interaction, $\chi^2(2) = 7.36$, $p = .025$. The planned high–low contrasts within each context included zero, and Study 1b provides the direct test of whether this secondary pattern recurs. The Bayesian predictive discriminability contrasts likewise showed no consistent direction across the three contexts. The central result is therefore a size-first preference graded by referential usefulness: it was strongest when size alone identified the target, weakest when colour alone did, and remained slightly positive in the colour-sufficient condition.

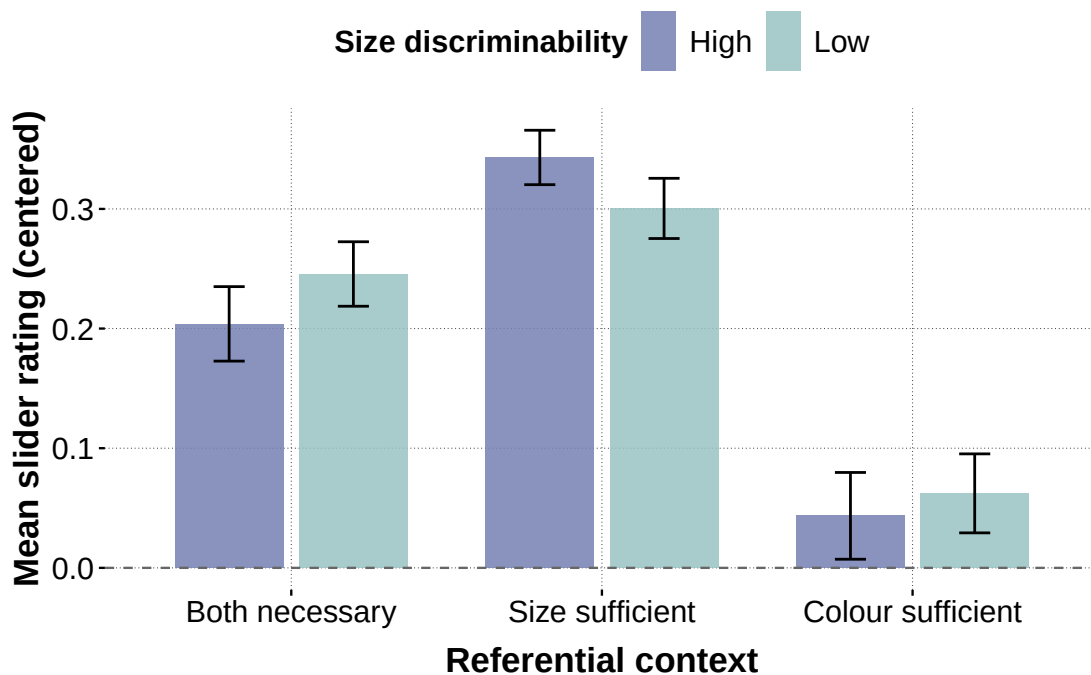


Figure 3

Slider-rating data. Mean centred slider ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.

3.3 Study 1b: Direct Slider Replication

3.3.1 Participants and preprocessing

The replication retained 104 of 132 participants after applying the Study 1a exclusions to an independent participant sample. The cleaned file contains the same 1,080 list–item–condition design tuples as Study 1a and 8,379 critical observations. The complete exclusion and preprocessing record appears in Appendix A.

3.3.2 Results

We fitted the same mixed-effects specification as in Study 1a to the replication ratings. Referential context was again significant, $\chi^2(2) = 38.651$, $p < .001$. Mean centred ratings were 0.272 in size-sufficient contexts, 0.197 when both properties were necessary, and 0.054 in colour-sufficient contexts. The corresponding 95% confidence intervals were [0.235, 0.309], [0.154, 0.240], and [−0.005, 0.113]. All three Bonferroni-corrected context

contrasts were significant. The colour-sufficient point estimate virtually reproduced Study 1a (.054 vs. .053), and both confidence intervals narrowly include zero. The matched Bayesian predictive analysis assigned posterior probability .999 to the graded context order. It estimated the colour-sufficient preference at .057, 95% credible interval [.001, .116], with posterior probability .977 of exceeding neutrality. Study 1b therefore directly replicates the primary context gradient and supplies convergent evidence for a small positive colour-sufficient preference. The weak Study 1a crossover did not recur, so size discriminability remains secondary to the replicated context pattern.

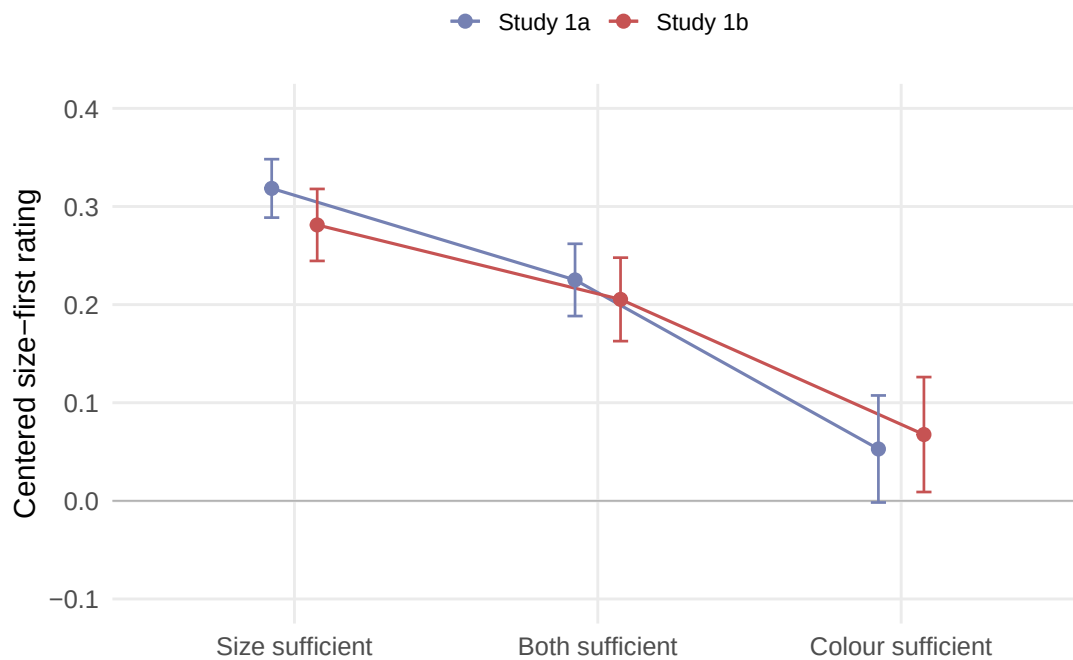


Figure 4

Combined Study 1a and Study 1b slider results. Mean centred ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.

3.4 Study 2: Production Study

3.4.1 *Participants*

Participants ($N = 113$ after exclusions; 60 in the LOW size discriminability group, 53 in the HIGH size discriminability group) were recruited online via Prolific; anyone who had completed either slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The canonical production table contains 9,100 annotated observations across all three adjective-pair families. The model comparison retains the native fifteen-category response space across size-colour, size-form, and colour-form trials. Adjective family is a structured dimension of the materials that specifies which two properties support reference; the primary experimental factors remain referential context and size discriminability. The targeted frequentist analysis uses size-colour to align directly with the slider studies, while size-form provides generalisation to a second size-containing family and colour-form provides a control for the full-data model comparison. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Study 1a.

3.4.2 *Procedure*

Study 2 used the shared referential design in a production format. Participants saw the framed target together with the fixed sentence frame *den* ____ *Aufkleber* and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot reduced off-task variability and provided a controlled basis for coding. The practice materials additionally introduced the relevant colour and form vocabulary before the critical trials began. Each participant completed 135 main trials: 81 critical trials and 54 filler or control trials from one of six counterbalanced lists.⁴

Responses were coded into 15 utterance categories corresponding to all ordered

⁴ The number of filler and control trials was reduced in Study 2 to keep the total number of trials manageable because typed production responses were slower and more effortful than slider judgments.

combinations of S (size), C (colour), and F (form): 3 one-adjective, 6 two-adjective, and 6 three-adjective strings. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label $y_i \in \{1, \dots, 15\}$ for each trial i . For supplementary comparison with the slider studies, we also tracked two derived binary projections: whether an utterance was size-initial and whether it contained more adjective material than the minimally sufficient property set. These projections combine several underlying choices and are not treated as direct measures of adjective order. For the mixed-effects logistic regressions, trials satisfying the coded outcome were coded as 1 and all other annotated trials were coded as 0.

3.4.3 *Results*

Figure 5 summarises all 9,100 production trials across size–colour, size–form, and colour–form. The complete categorical distribution shows that referential context affects which adjectives are selected and how much adjective material is produced. Both-necessary scenes generally elicit longer descriptions, whereas one-property-sufficient scenes more often elicit a single adjective. In particular, colour-sufficient size–colour scenes are commonly resolved with colour alone. On those trials, no size–colour ordering decision is observed.

Selection, adjective set, and length. Across families, speakers preferentially mentioned properties that distinguished the target in the current scene. The resulting unordered adjective-set distribution separates one-property responses from responses that contain both referentially relevant properties and from three-adjective descriptions. Lower size discriminability increased the probability of additional adjective material most clearly when size alone would otherwise have been sufficient. In the supplementary binary projection, over-informative responses were more frequent under low than high size discriminability in size-sufficient displays (.90 vs. .63). The Bayesian predictive difference was .234, 95% credible interval [.151, .318], and the targeted interaction against the other

contexts was .153, [.072, .236]. The visual manipulation therefore bears most clearly on adjective selection and utterance length rather than on a stable ordering preference.

Initial adjective and conditional order. Initial-adjective probabilities reflect both whether an adjective is selected and where it occurs. For this reason, size-initiality is a useful descriptive projection of the complete response space but not a direct counterpart of the slider outcome. Within the size–colour trials, the context-by-discriminability interaction for size-initial responses was significant, $\chi^2(2) = 11.7$, $p = .003$. High discriminability increased size-initial responses in both-necessary displays (.49 vs. .65), with a Bayesian predictive difference of .141, 95% credible interval [.036, .242]. The corresponding effects were small in the two one-property-sufficient contexts. Conditional on producing both size and colour, size-first descriptions remained more frequent than colour-first descriptions; colour-before-size reversals were comparatively rare. That conditional subset is selected by the prior decision to mention both properties and is therefore reported separately from adjective selection and length. The same distinction applies to size–form, colour–form, and three-adjective responses.

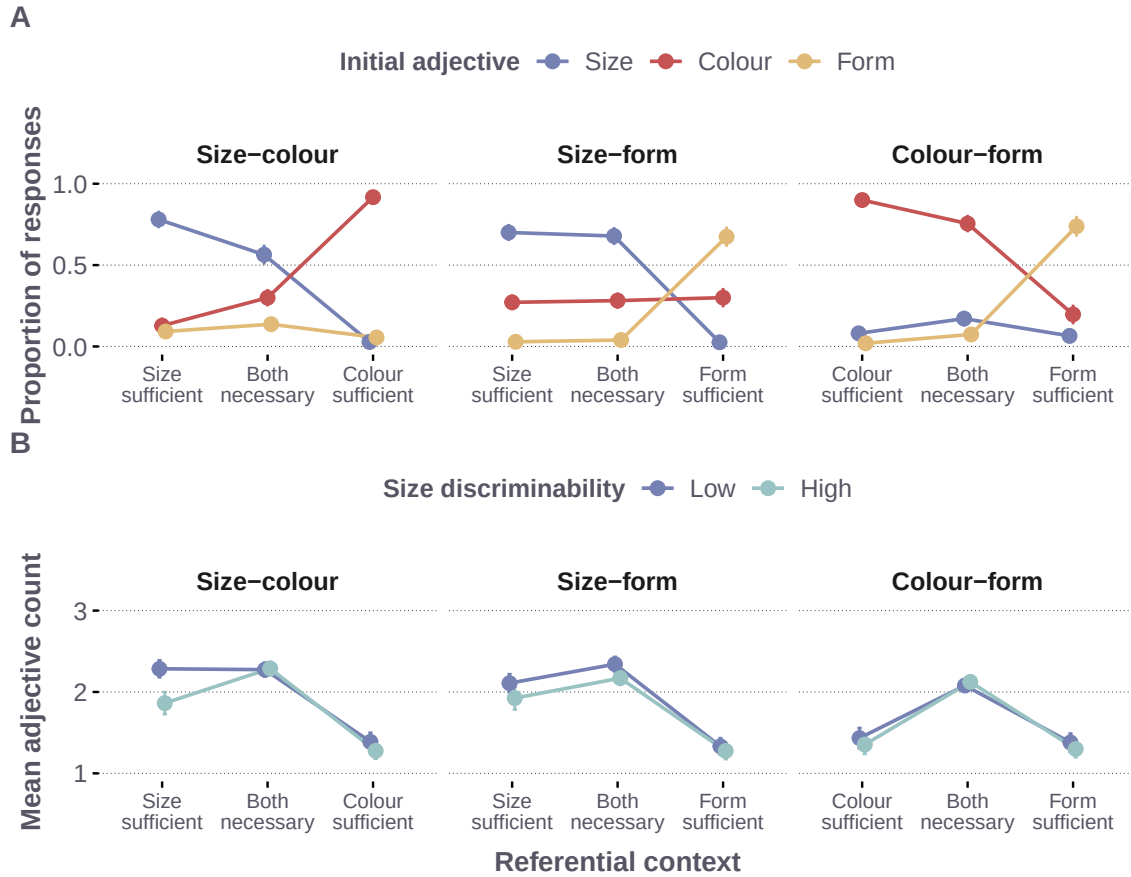
Replace the production figure with panels for adjective-set choice, response length, initial adjective, and conditional order within a fixed adjective set.

3.5 Behavioural Synthesis

The slider studies establish a replicated context gradient in explicit order preference. Ratings are highest when size identifies the target, lowest when colour does, and retain a small positive size-first estimate in the colour-sufficient condition. The size-discriminability interaction in Study 1a does not replicate and remains secondary.

Production exposes a different decision structure. Scene-specific usefulness affects adjective selection and utterance length, and one-adjective descriptions frequently resolve reference before an order among two adjectives arises. The size-discriminability manipulation primarily changes whether speakers add material, especially in size-sufficient scenes.

Initial-adjective summaries inherit both these selection effects and any subsequent ordering

**Figure 5**

Production data across all 9,100 annotated trials. The current panels show initial-adjective proportions by adjective family and referential context and mean adjective count by family, context, and size discriminability. The final figure will additionally distinguish unordered adjective-set choice from order conditional on a fixed adjective set. Error bars are 95% confidence intervals over participant-level summaries.

preference. The formal model must therefore predict the complete fifteen-category response distribution rather than an isolated size-first/colour-first choice.

4 A Model of Completed Referential Responses

The theoretical motivation distinguishes two directions of communicative efficiency: hierarchical semantic composition proceeds from noun-adjacent modifiers outwards, whereas the surface production path proceeds from the linearly first adjective onwards. The final

model keeps both directions explicit. The semantic path specifies how the meanings in a completed adjective sequence are composed from the syntactically inner modifier to the outer modifier; the production path specifies how candidate surface prefixes are evaluated from left to right. These are different computational orderings over the same linguistic material. We ask separately whether belief-dependent meaning makes the inner-to-outer semantic path predictively consequential in the present materials and how strongly prefix-specific competition along the surface path remains expressed in completed responses.

4.1 Response Space and Observed Display Representation

The response space contains all fifteen ordered strings formed without repeating adjective types from size (S), colour (C), and form (F): three one-adjective responses, six two-adjective responses, and six three-adjective responses. Each trial is represented by the raw observed six-object display. The final input is a deterministic 6×3 matrix containing observed size, colour, and form values, with the target at index 0. All visible properties are therefore available to the modelled listener; a family-masked representation is retained only as a robustness analysis.

4.2 Fixed Listener Semantics

Size is represented by a graded threshold over the scene-level comparison class $\mathcal{C}$ (Franke et al., 2019; Schmidt, Goodman, Barner & Tenenbaum, 2009):

$$\theta_{\delta}(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta \left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C}) \right), \quad (1)$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity.⁵ A perceptual-blur function makes size membership graded. Colour has a softened determinate meaning (Degen et al., 2020; Waldon & Degen, 2021). The selected F0 package treats form as neutral in the listener core and captures form production through the terminal response policy. This is the best tested predictive package, not an identified claim

⁵ The implementation records this threshold-selectivity constant as k ; the manuscript uses δ to distinguish it from the central prefix-normalisation parameter κ .

that participants assign form adjectives a neutral meaning.

The main listener uses context-fixed semantics: the size threshold is computed from the observed scene and remains constant while a surface prefix is evaluated. Belief-dependent size semantics is introduced separately in Section 6 as a predictive and simulation boundary analysis.

4.3 Hierarchical Inner-to-Outer Composition

The first computational direction follows the hierarchy of prenominal modification. Let a completed surface sequence be $u = w_1 \dots w_T$, and let $c_1, \dots, c_T$ denote the same modifiers ordered from syntactically inner to outer. For prenominal modification, the noun-adjacent adjective is therefore $c_1 = w_T$. Starting from the scene prior $H_0 = P_0$, hierarchical composition successively restricts the current listener state:

$$H_j(o) \propto H_{j-1}(o) \llbracket c_j \rrbracket_{\rho_j}(o), \quad j = 1, \dots, T. \quad (2)$$

For *big blue sticker*, the hierarchical path therefore incorporates *blue* before *big*, even though *big* is linearly first.

The semantic regime determines what the subscript ρ_j contributes. Under context-fixed semantics,

$$\llbracket c_j \rrbracket_{\rho_j}(o) = \llbracket c_j \rrbracket_{\text{fix}}(o), \quad (3)$$

so the meaning of size is anchored to the scene-level comparison class. Under belief-dependent semantics, the interpretation of a gradable size adjective can instead depend on the hierarchically prior listener state:

$$\llbracket S \rrbracket_{\rho_j}(o) = \llbracket S \rrbracket_{\theta_\delta(H_{j-1})}(o). \quad (4)$$

The noun-adjacent modifier can then change the comparison class against which an outer size adjective is interpreted.

This inner-to-outer order is a property of the semantic representation. It does not by itself assert that comprehenders process the noun-adjacent adjective earlier in real time.

With context-fixed meanings, the final literal restriction is order-insensitive because the same fixed meanings are multiplied together. The hierarchy becomes predictively consequential when a meaning depends on the intermediate state, as in the belief-dependent size account tested below and isolated in simulation.

4.4 Surface-Prefix Candidate Evaluation

The second computational direction follows the emerging surface sequence. For every candidate prefix, the listener distribution is reconstructed from the scene prior and each meaning in that prefix is applied once. Let $u_{\leq t} = w_1 \dots w_t$ be a left-to-right surface prefix. Under fixed semantics,

$$Q_t(o \mid u_{\leq t}) \propto P_0(o) \prod_{j=1}^t \llbracket w_j \rrbracket_{\text{fix}}(o). \quad (5)$$

Thus the production path for *big blue* evaluates a *big* prefix before considering *blue*, the reverse of the inner-to-outer semantic ordering for the completed phrase.

For the belief-dependent production variant, the same Variant-B factorisation reconstructs each candidate prefix from P_0 and applies every meaning exactly once, but a size meaning can be computed relative to the listener state produced by earlier *surface* material. This is deliberately distinct from Equation (2), where semantic dependency proceeds from the noun-adjacent modifier outwards. The distinction between these two update directions is one of the paper’s central theoretical contrasts.

Variant B replaces an earlier repeated-evidence implementation in which selected prefix meanings entered both the carried state and the candidate score. The revised factorisation changes neither the existence nor the direction of hierarchical semantic composition; it only ensures that the surface-prefix production calculation does not count previously selected evidence twice.

4.5 Raw Path Utilities

At prefix $u_{< t}$, the raw utility of candidate adjective a for target o^* is

$$r_{it}(a; u_{< t}, o^*) = \alpha_i \log Q_t(o^* \mid u_{< t} + a) + \lambda_{\text{salience}} z_a, \quad (6)$$

where α_i is participant-specific response determinism and z_a is a centred display-derived salience score. The completed path accumulates the utilities of selected adjectives:

$$R_i(u, o^*) = \sum_t r_{it}(w_t; u_{<t}, o^*). \quad (7)$$

For the same path, the accumulated prefix-specific log normalisers are

$$N_i(u, o^*) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp r_{it}(a; u_{<t}, o^*), \quad (8)$$

where $\mathcal{A}(u_{<t})$ contains adjective types not yet used in the prefix. Stopping is not a locally normalised action in the selected model. Utterance length is represented in the terminal response policy described below.

4.6 Stable Order Expectations and Terminal Response Policy

Stable order expectations enter through a German language model’s centred log-probability residual within each unordered adjective set. Let $\Omega(u)$ denote this residual. It distinguishes permutations while summing to zero within each adjective set. Its weight varies hierarchically across participants:

$$\log \beta_i = \log \beta_0 + \tau_\beta b_i, \quad b_i \sim \mathcal{N}(0, 1). \quad (9)$$

This hierarchy controls substantial participant variation in stable order weighting. It is an auxiliary random effect rather than a theory of the linguistic origin of canonical adjective order.

The terminal response policy $T(u, X)$ links the shared pragmatic path to the task’s complete response distribution. It includes a sufficient-one-word term, form backup, a three-word penalty, visual size-reliability terms, and the continuation cost associated with display salience. These components are shared across the compared architectures. They model adjective-set selection and length beyond what follows from target-directed path utility alone. The registered POLICY-LIGHT analysis shows that the relative preference for a graded architecture survives removal of the main direct length and backup terms, while absolute predictive accuracy deteriorates substantially. The successful account is therefore a probability architecture embedded in a task-specific terminal policy.

4.7 Global, Graded, and Fully Prefix-Normalised Architectures

The three final speakers share R_i , N_i , Ω , and T and differ only in the weight assigned to accumulated prefix normalisers:

$$S_{\kappa,i}(u \mid o^*, X) \propto \exp\{R_i(u, o^*) - \kappa N_i(u, o^*) + \beta_i \Omega(u) + T(u, X)\}. \quad (10)$$

The globally normalised endpoint G-HO fixes $\kappa = 0$. The graded model K-HO estimates one shared population value $0 < \kappa < 1$. The fully prefix-normalised endpoint I-HO fixes $\kappa = 1$. At common parameter values, the graded implementation recovers both endpoints exactly.

The parameter κ is an effective weight on the contribution of prefix-specific competition to the completed-response distribution. It is conditional on the listener representation, stable order model, and terminal response policy. An interior value does not by itself show that speakers temporally commit to a particular number of words in advance or plan one token at a time.

4.8 Theoretical Core and Task Links

4.9 Inference and Predictive Estimand

Models were implemented in NumPyro (Phan, Pradhan & Jankowiak, 2019) and estimated with four NUTS chains (Hoffman & Gelman, 2014), each using 1,000 warm-up iterations and 1,000 retained draws. The likelihood is categorical over all fifteen completed responses for each of the 9,100 production trials. Predictive adequacy was compared with observation-level PSIS-LOO (Vehtari, Gelman & Gabry, 2017); uncertainty for the central contrasts was additionally aggregated and bootstrapped at the participant level. All final models use the same response support and matched auxiliary components. The estimand is the probability of a held-out completed response from the sampled task population. It is not a direct estimate of online planning time.

Table 1*Theoretical core and auxiliary task links in the final production model.*

Component	Classification	Role and evidential interpretation
Raw path utility R_i	Core	Target-directed utility accumulated along surface prefixes
Prefix normalisers N_i and κ	Core	Competing completed-response probability architectures
Fixed listener semantics	Shared representation	Common interpretation used in the final architecture comparison
LM residual $\Omega(u)$	Stable-order task link	Imports relative order expectations within adjective sets
Participant order weight β_i	Auxiliary hierarchy	Controls participant variation in stable order expectations
Salience	Task link	Display-derived prominence in candidate utility
One-word, backup, three-word, and reliability terms	Terminal task link	Supports adjective-set and utterance-length prediction
Participant response determinism α_i	Repeated-response hierarchy	Controls participant variation in choice concentration

5 Population Architecture Results

5.1 Final Matched Model Comparison

The final comparison controls participant-specific variation in the LM-derived stable order weight in all three architectures. The globally normalised endpoint G-HO, the graded model K-HO, and the fully prefix-normalised endpoint I-HO otherwise share the observed input, fixed semantics, terminal response policy, participant rationality hierarchy, and response support. All three fits had zero divergences, maximum $\hat{R} \leq 1.006$, and no observation with Pareto $k > .7$.

K-HO exceeded G-HO by 229.605 ELPD units. The participant-clustered total

Table 2

Final nuisance-controlled production-model comparison. Higher ELPD indicates better out-of-sample prediction.

Model	ELPD	Δ from K-HO	Max $\hat{R}$	Max Pareto k
G-HO	-13,771.122	-229.605	1.0055	.466
K-HO	-13,541.517	0	1.0046	.461
I-HO	-13,963.904	-422.387	1.0049	.522

standard error was 81.097, and a participant bootstrap gave a 95% interval of [58.765, 376.160]. The participant-level win rate was .717. The globally normalised endpoint therefore underexpresses the predictive consequences of competition within successive prefix-specific alternative sets.

K-HO exceeded I-HO by 422.387 ELPD units. The participant-clustered total standard error was 108.714, and the participant-bootstrap 95% interval was [221.275, 647.382]. The participant-level win rate was .637. Retaining the full prefix-normaliser contribution is therefore also too strong for the completed-response distribution. The difference between the two endpoints was comparatively uncertain once participant order variation was controlled.

5.2 Population Prefix-Normalisation Weight

The shared population weight was clearly interior:

$$\kappa = .4472, \quad 95\% \text{ CI} = [.4036, .4897]. \quad (11)$$

The posterior excludes both endpoint regions under the final specification. The precise value is nevertheless model-conditional: it changes when the listener’s feature scope or auxiliary policy changes. The robust conclusion is that an interior contribution of prefix-specific normalisers predicts better than either endpoint, not that human speakers implement a fixed proportion of incremental planning.

5.3 Where the Architecture Gain Occurs

The architecture advantage is not confined to the order of two already-selected adjectives. Relative to G-HO, positive K-HO contributions are largest in colour–form and size–colour trials, in two- and three-adjective responses, in the colour–form and three-adjective sets, and among colour-initial responses. Relative to I-HO, the largest gains occur in both size-containing families, in one- and three-adjective responses, and across several adjective sets and initial-adjective categories. Selected positive contributions are shown in Table 3. These decompositions localise predictive gains and losses; the cells are not independent hypothesis tests.

Table 3

Selected positive ELPD contributions to the final architecture contrasts.

Dimension	K-HO minus G-HO	Contribution	K-HO minus I-HO	Contribution
Family	Colour–form	140.953	Size–colour	256.543
	Size–colour	81.461	Size–form	209.579
Length	Two adjectives	118.682	One adjective	235.955
	Three adjectives	104.183	Three adjectives	144.046
Adjective set	$\{C, F\}$	245.082	$\{C\}$	151.385
	$\{S, C, F\}$	104.183	$\{S, C, F\}$	144.046
Initial adjective	Colour	439.253	Form	181.764

This distribution of gains supports the paper’s central behavioural premise. The normalisation architecture predicts the joint response space over adjective selection, length, set, and initial adjective. It is not merely a flexible parameter for the conditional order of a fixed pair.

5.4 Dependence on the Terminal Response Policy

A registered POLICY-LIGHT comparison removed the sufficient-one-word bonus, form backup, three-word penalty, and size-reliability terms as one bundle. The graded model

continued to outperform both endpoint architectures and retained an interior κ . All three reduced-policy models, however, lost substantial absolute predictive accuracy relative to their FULL counterparts. The architecture preference is therefore not created entirely by the direct length and backup terms, but accurate prediction of the complete response distribution requires the terminal task link.

5.5 Participant-Level Identification Audit

Allowing participant-specific prefix-normalisation improved prediction relative to the shared model and to a matched participant-specific length-bias model. An equally flexible participant-specific stable-order model performed indistinguishably, and the participant- κ fit also failed the registered diagnostic and dependence gates. We therefore do not interpret individual κ_i values as identified speaker profiles. The full hierarchy comparison and descriptive parameter-decoding summaries appear in Appendix E. Participant-specific stable-order weighting is retained in the final population models as an auxiliary random effect because the matched G-HO/K-HO/I-HO comparison shows that the shared architecture result survives that control.

6 Semantic Updating as a Boundary Case

6.1 Predictive Equivalence in the Production Data

The two directional accounts of communicative efficiency motivate a separate test of whether the inner-to-outer semantic path becomes order-sensitive when size meaning depends on the intermediate comparison class. The formal hierarchy in Equation (2) remains part of the theory under both semantic regimes. What differs is whether the interpretation of size is fixed from the scene or recomputed from the hierarchically prior state.

In the matched deterministic-input comparison, belief-dependent updating did not improve prediction: updating minus fixed semantics was -0.26 ELPD, with participant-bootstrap interval $[-0.97, 0.44]$. Under the production constants and actual size-colour displays, recomputing the size threshold after a colour restriction changed the target’s graded size value and the literal listener’s target probability by less than .001 on

average. The fitted materials therefore place the model in a regime where making the hierarchical composition path belief-dependent is an extremely small perturbation.

This null result does not remove inner-to-outer composition from the account. It limits a more specific claim: the present production data do not require the comparison class for size to be dynamically recomputed along that hierarchy once the completed-response architecture and task links are modelled. We use simulation to ask a different question—whether belief-dependent hierarchical composition can generate an order asymmetry under conditions in which its semantic consequences are larger.

6.2 When Can Hierarchical Updating Create an Order Asymmetry?

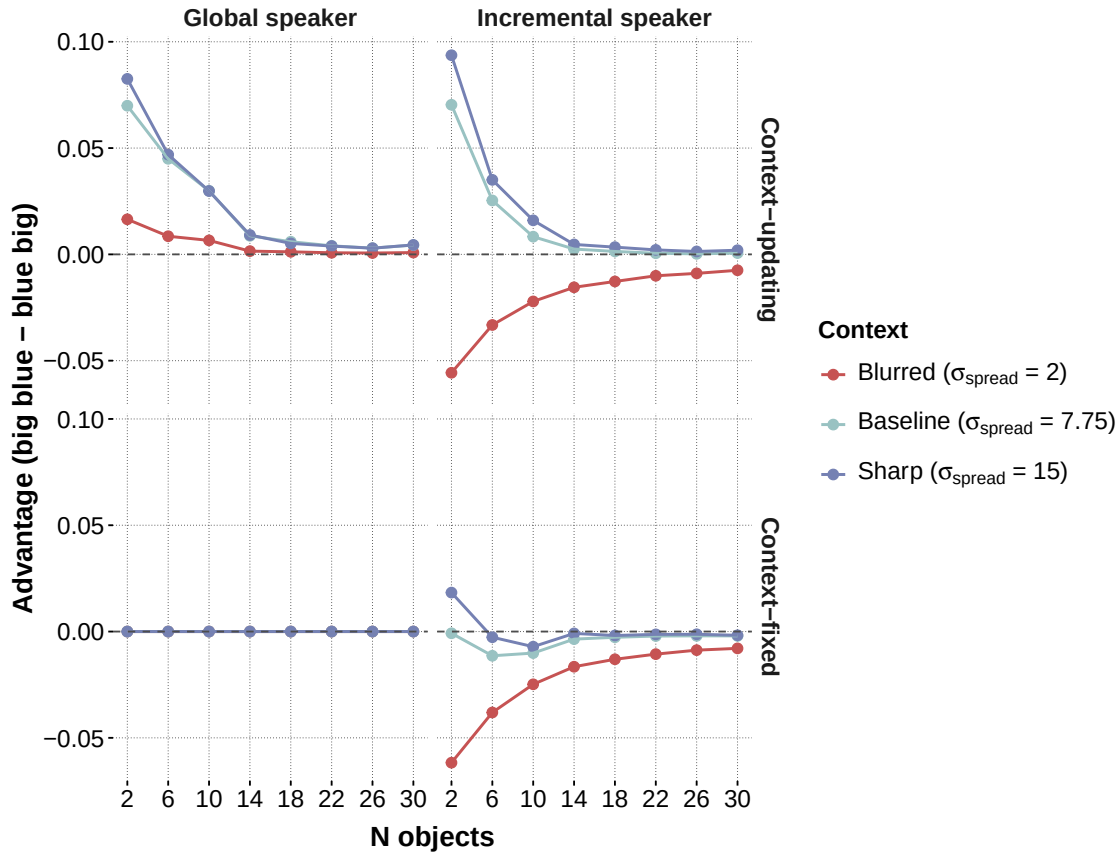
The simulation removes the fitted LM order residual, visual-salience calibration, terminal policy, and participant variation. Both surface orders are interpreted by a hierarchically compositional listener that combines the noun-adjacent adjective first. We cross global versus prefix-normalised speaker architectures with fixed versus belief-dependent size semantics over a broad grid of context sizes, size distributions, colour reliability, threshold selectivity, and perceptual blur. Full grid details appear in Appendix H.

For each simulated context, we compute the pragmatic listener’s target probability after *big blue* and after *blue big*. The size-first advantage is

$$\Delta_{\text{SF}} = L_1(o^* \mid \text{“big blue”}) - L_1(o^* \mid \text{“blue big”}). \quad (12)$$

Positive values indicate a higher model-assigned target probability for the size-first order.

Belief-dependent hierarchical updating produces a positive size-first advantage across much of the high-reliability simulation grid, including under a globally normalised speaker. The corresponding fixed-semantic global model assigns the same target probability to the two orders. Prefix normalisation changes the shape of the asymmetry and can yield a colour-first advantage when size is weak and the context is small. The simulation therefore establishes generative sufficiency and boundary conditions. It does not show that the semantic update contributed to the fitted production responses.

**Figure 6**

Boundary simulation of the size-first advantage Δ_{SF} as a function of context size and contextual spread, faceted by speaker architecture and semantic regime. Positive values indicate a higher model-assigned target probability for the size-first order. The figure characterises model behaviour under simulated assumptions; it does not identify the mechanism used by participants.

7 General Discussion

The behavioural studies show why adjective ordering cannot be isolated from the broader problem of reference generation. Explicit slider judgements exhibit a replicated context gradient: size-first preference is strongest when size identifies the target, weakest when colour does, and remains slightly positive in colour-sufficient scenes. Production distributes the same contextual pressure across a larger response space. Speakers often select one sufficient adjective, and visual size reliability primarily affects whether they add material. On those trials, selection and utterance length determine whether a multi-adjective ordering decision arises at all.

The final production comparison evaluates that complete response space. After participant-specific variation in stable LM-derived order expectations is controlled, the shared graded model outpredicts both a globally normalised endpoint and a fully prefix-normalised endpoint. The estimated prefix-normalisation weight is clearly interior, and the predictive gains extend across adjective families, response lengths, adjective sets, and initial adjectives. This is the paper’s central computational result. It concerns the predictive consequences of different completed-response probability architectures, not the temporal implementation of human planning.

The final model also clarifies what the architecture result does not explain by itself. Stable order expectations enter through a German language-model residual whose influence varies across participants. The model therefore controls, rather than derives, the conventional size-first tendency. Likewise, accurate prediction of adjective selection and length requires a terminal response policy containing salience, one-word support, form backup, and reliability-related terms. A reduced policy preserves the relative preference for an interior architecture but loses substantial absolute predictive accuracy. The successful account is consequently a graded normalisation architecture embedded in an explicit task link.

The two directional accounts of communicative efficiency remain theoretically useful, but the fitted evidence is asymmetric. Belief-dependent size updating makes no practically

meaningful predictive contribution in the present materials. Controlled simulation shows that hierarchical updating can create a size-first target-probability advantage when colour is more reliable and the semantic update is less diffuse. That result establishes a possible semantic route under the implemented assumptions. It does not show that participants used the route in this experiment, and it does not restore an empirically identified division of labour between semantics and production.

The participant hierarchy yields a second negative identification result. Participant-specific prefix-normalisation improves prediction relative to a shared model, but participant-specific stable-order weighting performs indistinguishably. The completed responses therefore reveal systematic participant consistency without identifying its source as an individual prefix-normalisation profile. The final model retains participant order variation as an auxiliary control and one shared population κ . Individual κ_i values are not interpreted as stable strategies, planning traits, or cross-session characteristics.

The distinction between hierarchical semantic computation and surface-prefix probability architecture should also remain explicit. Hierarchical composition specifies which semantic restriction is computed inside another in a formal representation. The production model evaluates surface prefixes and retains a graded contribution from their normalisers in the completed-response distribution. Neither operation, as measured here, establishes the millisecond time course of semantic interpretation or utterance planning. Online measures would be needed to identify temporal implementation.

The experiments use German prenominal descriptions in a controlled reference task with a fixed adjective vocabulary and a restricted fifteen-response support. The final form package is predictive rather than an identified theory of form meaning, and the stable order component is imported rather than explained. Further work can test whether the architecture generalises to freer production, new lexical families, different adjective systems, and designs that independently manipulate lexical subjectivity and perceptual reliability. Such extensions should preserve the distinction between response-space prediction, semantic

sufficiency, and process identification.

The strongest conclusion is therefore bounded but substantive. Adjective order is an outcome of choices about what to mention and how much to say. Within that response-generation problem, completed descriptions are predicted better by an interior contribution of prefix-specific normalisers than by either endpoint architecture, even after substantial participant variation in stable order expectations is controlled.

8 Data Availability Statement

Data, analysis scripts, model code, final posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the deterministic raw-observed production input, the final G-HO/K-HO/I-HO artifacts, preprocessing and reconstruction receipts, and the scripts required to reproduce the reported behavioural and predictive analyses.

9 Author Note

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Appendix A

Methods and Coding Transparency

Table A1 summarises the participant exclusions and the analysis tables used in the reported manuscript analyses. The per-criterion exclusion counts are not mutually exclusive, so the unique number of excluded participants is smaller than the sum of the criterion counts. For Study 2, the canonical production table contains all three adjective-pair families. The production model comparison reported in Section 5 uses all 9,100 canonical annotated observations over the 15 utterance categories and three adjective-family pairings. The size–colour marginal is retained as the theoretical anchor and descriptive subset.

Table A1

Participant exclusions and analysis-table sizes. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Analysis table used in manuscript
Study 1a slider ratings	150 $\rightarrow$ 134	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three “none fits” responses on experimental items; 16 unique participants excluded.	Broad cleaned critical table: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	132 $\rightarrow$ 104	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	Cleaned replication table: 8,379 critical observations. Exact design-equivalent tuples: 1,080.
Production study	149 $\rightarrow$ 113	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Canonical full-data table: 9,100 annotated observations. All three adjective-family pairings enter the final production-model comparison.

The slider-rating materials used the fixed adjective vocabulary in Table A2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table A2

Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels 1–6, 9, 10	<i>großen, kleinen</i>
Colour	blue, orange, grey, brown, green, black	<i>blauen, orangen, grauen, braunen, grünen, schwarzen</i>
Form	circle, diamond, heart, quadrat, star, triangle	<i>runden, diamantförmigen, herzförmigen, quadratischen, sternförmigen, dreieckigen</i>

The final modelling table is reconstructed deterministically from the raw stimulus records. Observed colour and form values are recoded into target–object relations without stochastic nuisance-feature generation; this recoding is separate from the lexical forms shown to participants.

Production responses were coded into the 15 ordered adjective categories listed in Table A3. The data files use D for the size or dimension adjective. The model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size expressions, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table A3

Production coding dictionary. In the data files, D corresponds to the size adjective S ; C and F correspond to colour and form.

Length	Codes
One adjective	D, C, F
Two adjectives	DC, DF, CD, CF, FD, FC
Three adjectives	DCF, DFC, CDF, CFD, FDC, FCD

The behavioural hypotheses were evaluated through parallel frequentist mixed-effects tests and Bayesian predictive contrasts. Both analyses used the same fixed-effects design and random-effects structure. The later Bayesian production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete response space. For the slider analysis, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. The reported linear mixed-effects model included fixed effects of referential context, size discriminability, and their interaction, plus participant intercepts and context slopes:

$$y_{\text{slider}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}).$$

Adding an item intercept yielded an estimated variance of zero and a singular fit. The reported participant-slope model converged without a singularity. In Study 1b, the context-by-discriminability interaction was small, $\chi^2(2) = 1.178$, $p = .555$. For production size-initiality, the logistic mixed-effects model used a binary outcome indicating whether the coded response began with the size adjective:

$$y_{\text{size-initial}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}) + (1 \mid \text{item}).$$

For production over-informativeness, the descriptive model used the same fixed and random structure with a binary outcome indicating a three-adjective response in both-necessary

contexts or a multi-adjective response in one-property-sufficient contexts:

$$y_{\text{over-informative}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}) + (1 \mid \text{item}).$$

The two production models converged without singular fits. For size-initiality, the planned discriminability contrast was supported in the both-necessary context ($p = .004$) and uncertain in the size- and colour-sufficient contexts ($p = .36$ and $p = .60$). For over-informativeness, the corresponding contrast was supported in the size-sufficient context ($p < .001$) and uncertain in the other two contexts ($p = .43$ and $p = .11$). The analysis script `paper/scripts/behavioural_robustness.R` writes the reported marginal estimates, planned contrasts, likelihood-ratio tests, and convergence checks to four CSV files under `paper/data/`.

The complementary Bayesian analyses retained these outcome definitions and random-effects structures. The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, $\text{Normal}(0, .30)$ priors on fixed effects, $\text{HalfNormal}(.20)$ priors on participant standard deviations, and a $\text{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\text{Normal}(0, 2)$ intercept prior, $\text{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\text{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\text{LKJ}(2)$ prior. Four vectorized chains each provided 1,000 retained draws after 1,000 warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\hat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned predictive hypotheses were then evaluated on the rating or response-probability scale. The script `analysis/behavioural_bayes.py` writes the cell predictions, planned hypothesis contrasts, and inference diagnostics to `paper/data/behavioural_bayesian_predictions.csv`, `paper/data/behavioural_bayesian_hypotheses.csv`, and

`paper/data/behavioural_bayesian_diagnostics.csv.`

Appendix B

Final Production Model Specification

10.1 Deterministic Input Reconstruction

The final production input is reconstructed one-to-one from the raw stimulus table using participant list, condition, and item identifiers. Each retained response is paired with one observed 6×3 display tensor containing size, colour, and form values. An earlier preprocessing pipeline generated nuisance-feature values stochastically for some nominally irrelevant dimensions; the final analysis removes this transformation. Legacy first-draw, last-draw, and agreement-subset analyses are retained only as reproducibility sensitivities. The architecture and semantic-boundary conclusions are qualitatively stable across those legacy specifications.

10.2 Component Inventory

Table B1

Final production-model components.

Component	Status	Role
Observed E0 input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Soft determinate colour meaning
Form reliability	Fixed neutral	F0 listener package; form production supported by terminal backup
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
LM order residual	Fixed feature	Stable relative order score within adjective sets
LM order weight	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Terminal policy coefficients	Estimated	One-word support, form backup, length and reliability terms
Prefix-normalisation weight κ	Estimated in K-HO	Shared architecture parameter

Insert the frozen prior inventory for the final K-HO model from the v10 model registry.

Appendix C

Final Architecture Comparison and Diagnostics

Table C1

Final architecture contrasts after participant-specific stable-order variation is controlled.

Contrast	ΔELPD	Paired SE	Participant-clustered SE	Bootstrap 95% interval
K-HO minus G-HO	229.605	20.900	81.097	[58.765, 376.160]
K-HO minus I-HO	422.387	35.615	108.714	[221.275, 647.382]
I-HO minus G-HO	-192.782	—	169.152	[-553.149, 108.336]

All three final models used four chains, 1,000 warm-up iterations, and 1,000 retained draws. They had zero divergences, maximum $\hat{R} \leq 1.006$, minimum BFMI above .64, and no observation with Pareto $k > .7$. The exact endpoint audit returned maximum probability and log-probability differences of zero when the graded implementation was evaluated at $\kappa = 0$ and $\kappa = 1$.

Appendix D

Auxiliary Form, Order, and Policy Comparisons

Form package. The selected F0 package fixes form reliability at .50 and estimates a terminal form-backup coefficient. Two fixed informative alternatives tied form reliability to the colour value, with either the original or a strongly shrunk backup prior. Neither alternative was predictively equivalent to F0 in the full response space. This comparison supports F0 as an auxiliary predictive specification but does not identify participants' lexical semantics for form.

Stable order source. The selected O0 package uses the German LM residual alone. Adding a reliability-order term yielded worse conditional-order prediction within fixed adjective sets. Models that removed the LM residual did not provide valid posterior or PSIS evidence. The comparison therefore rejects the proposed addition but does not establish the LM residual as a unique cognitive source of order expectations.

Terminal policy. POLICY-LIGHT removed the sufficient-one-word bonus, form backup, three-word penalty, and size-reliability terms as one bundle. The graded architecture retained its relative advantage over both endpoints, while all reduced-policy models lost substantial absolute predictive accuracy.

Insert the compact F0/FT/FTS, O0/O1, and POLICY-LIGHT numerical tables from the final analysis receipts.

Appendix E

Participant Hierarchy Audit

The participant hierarchy audit compared the shared- κ model (H0) with participant-specific prefix-normalisation (HK), a participant-specific generic length bias (HL), and participant-specific LM-order weighting (HO). The registered contrasts were:

$$\text{HK} - \text{H0} = +317.301 \text{ ELPD},$$

$$\text{HK} - \text{HL} = +317.891 \text{ ELPD},$$

$$\text{HK} - \text{HO} = -2.184 \text{ ELPD}.$$

The HK–HO participant-bootstrap interval was $[-154.051, 137.077]$. HK also had maximum $\hat{R} = 1.0133$, two observations with Pareto $k > .7$, and a failed participant-scale dependence screen. The audit therefore supplies no evidence that participant-specific κ_i is uniquely identified relative to stable-order heterogeneity. Descriptive correlations between fitted κ_i values and the same participants’ response summaries decode the latent parameter within that fit; they are not independent validation.

Appendix F

Endpoint Identity and Architecture Audit

At common parameter values, the graded K-HO implementation with $\kappa = 0$ exactly recovers G-HO, and with $\kappa = 1$ exactly recovers I-HO. The participant-specific order mapping, terminal response-policy kernel, and response support are identical across the three architectures. Maximum category-probability and log-probability differences were zero at a numerical tolerance of 10^{-10} .

Appendix G

Belief-Dependent Size Semantics

For the hierarchical semantic path, let H_{j-1} be the state after the syntactically inner modifiers have composed. The updating listener recomputes the size threshold from this hierarchically prior belief state:

$$\llbracket S \rrbracket_{\text{upd}, H_{j-1}}(o) = \llbracket S \rrbracket_{\theta_\delta(H_{j-1})}(o), \quad (\text{G1})$$

whereas the fixed listener anchors the threshold to the scene prior:

$$\llbracket S \rrbracket_{\text{fix}}(o) = \llbracket S \rrbracket_{\theta_\delta(P_0)}(o). \quad (\text{G2})$$

For *big blue*, hierarchical composition applies the colour restriction before interpreting size. The distinct surface-prefix production calculation follows left-to-right order and, under Variant B, reconstructs each candidate prefix from the prior while applying every meaning once. The matched E0 comparison yielded no practically meaningful predictive difference between fixed and updating semantics.

Appendix H**Simulation Parameter Sweeps**

Figures H1 to H3 show how the three fixed semantic parameters affect communicative success across the two speaker architectures.

Table H1

Simulation parameter grid

Parameter	Grid values	# levels
Speaker architecture	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν	0.90, 0.92, 0.94, 0.96, 0.98	5
k	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination	1,000	
Total trials	12,000,000	

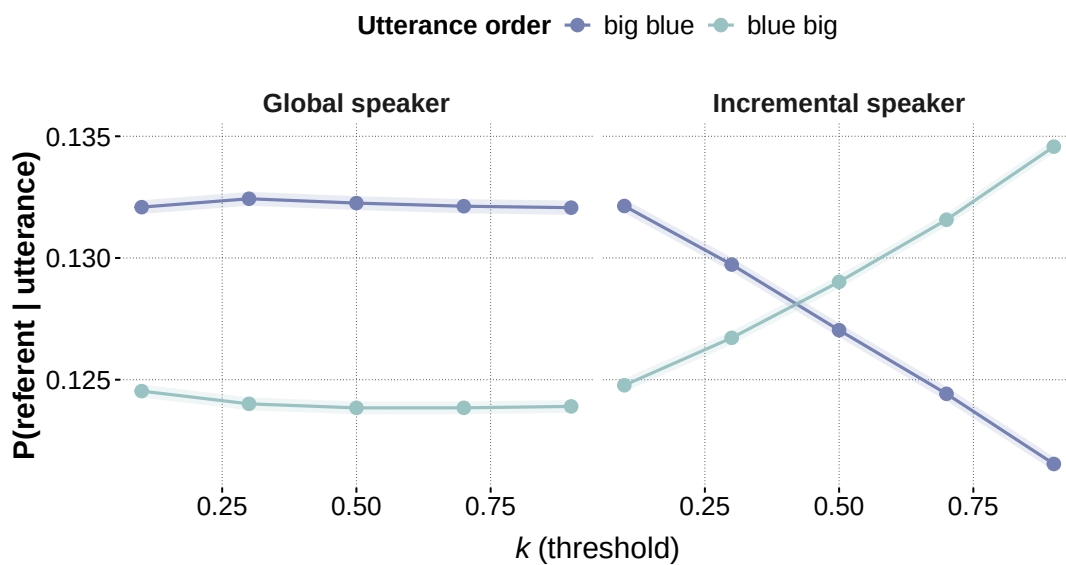


Figure H1

Effect of the threshold parameter k on communicative success in the simulation.

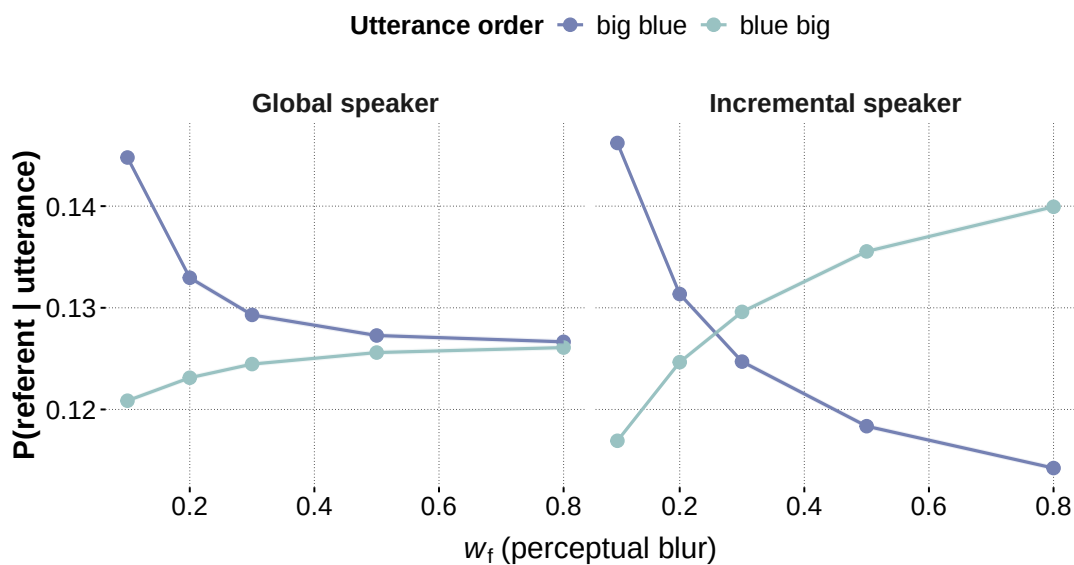


Figure H2

Effect of perceptual blur w_f on communicative success in the simulation.

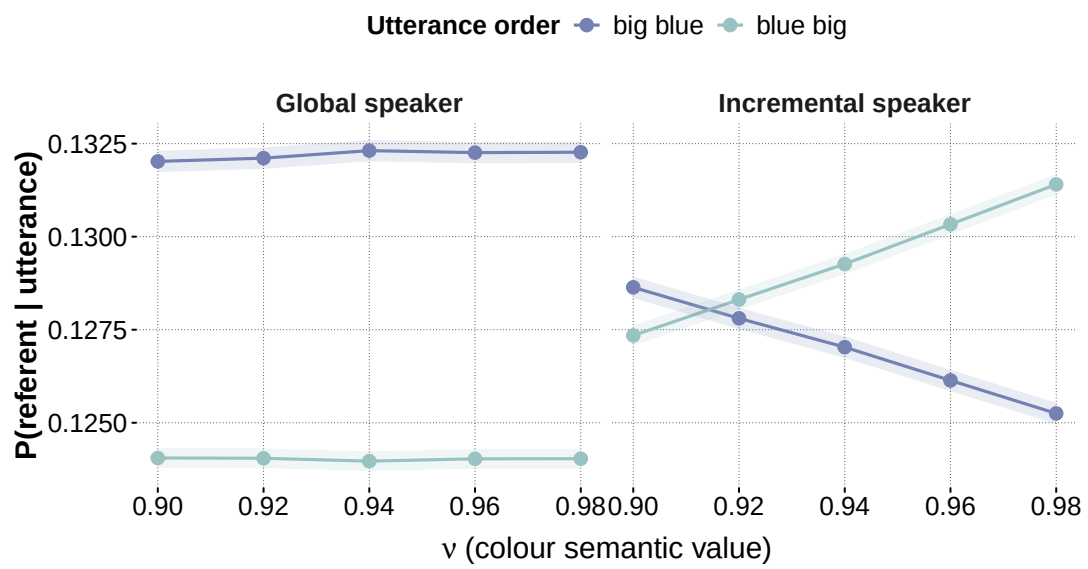


Figure H3

Effect of the colour semantic value ν on communicative success in the simulation.

Appendix I

Slider Colour-Reliability Audit

The slider-rating model fixes colour reliability ν at 0.80 in its diagnostic specification. Production uses the softer colour value $\nu_{\text{colour}} = 0.59$ in the production speaker (Section 5); for parity, we apply the same kind of sensitivity audit to the slider’s incremental speakers. The audit samples $\nu \sim \text{Uniform}(0.5, 0.999)$ within the model and recomputes the literal listeners at each MCMC step. Every other component of the incremental speaker remains unchanged. Table I1 reports the resulting posteriors alongside the originally fixed- ν fits.

Table I1

Slider colour-reliability audit. Fixed- ν rows use $\nu = 0.80$; free- ν rows sample $\nu \sim \text{Uniform}(0.5, 0.999)$ and recompute literal listeners each step. “ELPD” is PSIS-LOO; ELPD loss and dse are relative to the incremental, context-updating fixed- ν model. All four models use 4 chains $\times$ 500 warmup $\times$ 500 samples; maximum split- $\hat{R}$ rounds to 1.01 for the free- ν models.

Model	ELPD	ELPD loss	dse	α (sd)	b (sd)	ν (94% HDI)
Incremental, updating, fixed $\nu = 0.80$	-2,774.1	0.0	—	0.69 (0.06)	0.58 (0.06)	—
Incremental, fixed, fixed $\nu = 0.80$	-2,774.1	0.06	0.18	0.69 (0.06)	0.58 (0.06)	—
Incremental, updating, free ν	-2,775.0	0.89	1.08	1.53 (0.61)	0.41 (0.15)	0.63 [0.50, 0.82]
Incremental, fixed, free ν	-2,774.3	0.23	1.00	1.48 (0.55)	0.42 (0.13)	0.63 [0.51, 0.79]

Three points stand out. First, ν is weakly identified in the slider data: the updating credible interval includes the fixed value 0.80, the fixed-semantics credible interval reaches 0.79, and both posteriors place substantial mass near the lower prior boundary at 0.5. The production estimate has a narrower credible interval, 0.59 [0.56, 0.62]. Second, the nominal PSIS-LOO losses are all smaller than their dse (loss/dse < 1 throughout), and the shared

PSIS warnings limit these values to diagnostic use. Third, under the freed posterior the rationality parameter α moves to roughly 1.5 (from 0.69) and the bias intercept b moves below 0.5 (from 0.58), because ν and the other parameters trade off: a softer colour predicate forces a stronger rationality term and reduces the residual size-first bias absorbed by b . We therefore retain the fixed $\nu = 0.80$ for the slider model in the main text and treat α and b at that specification as the reported quantities. The fixed- ν estimate $b = 0.58$ remains conditional on this specification, and its archived credible interval includes 0.5. The broader conclusion is unchanged: the slider response space weakly identifies semantic scale parameters, and the fitted PSIS-LOO comparisons remain diagnostic.